

Honeybee dominance reshapes floral resource use of wild pollinators across Europe

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ABSTRACT

Apis mellifera L. (the western honeybee) is one of the most abundant and frequent pollinator species occurring in plant-pollinator networks worldwide. Its high abundance is often related to the introduction of managed colonies for honey production and crop pollination, raising concerns about potential impacts on wild pollinators due to resource competition and pathogen spill-over. By using a large plant-pollinator interaction dataset, we quantified the resource overlap of wild pollinators with the honeybee and their diet specialisation along a gradient of honeybee dominance across Europe. Honeybee dominance reduced resource use overlap between honeybees and wild pollinators indicating strong floral niche partitioning, yet wild pollinators became more generalist, visiting a greater diversity of floral resources. Using location-scale models, we observed a more inconsistent and unpredictable behaviour across wild pollinators when honeybees were very abundant. Increasing local floral diversity provided foraging opportunities for wild pollinators, but did not mitigate these effects. Our results suggest that avoidance and interference behaviours between managed and wild pollinators are common across Europe, but the importance of these resource use shifts for pollinator population dynamics, community diversity changes, and ecosystem functioning in the long term remains an open question.

INTRODUCTION

Apis mellifera L. (the western honeybee) is one of the most abundant and frequent pollinator species occurring in plant-pollinator networks worldwide across both agricultural and natural landscapes (Hung et al. 2018). Its high abundance is often related to the introduction of managed colonies for honey production and crop pollination (Gazzea, Batary, and Marini 2023; Visick and Ratnieks 2023), raising concerns about potential impacts on wild pollinators due to competition for floral resources (Iwasaki and Hogendoorn 2022; D. M. Thomson and Page 2020) and pathogens' spillover (Maurer et al. 2024). Field studies manipulating colony density in the short term have shown that honeybees can often cause competitive displacement through the use of shared resources (Hudewenz and Klein 2015; Magrach, Tobajas, and Martin 2025; Magrach et al. 2017; Pasquali et al. 2025; Valido, Rodríguez-Rodríguez, and Jordano 2019), forcing other pollinators to change their visitation patterns to alternative floral resources that are less preferred by the honeybee (Page and Williams 2023; Pasquali et al. 2025; D. M. Thomson 2016; D. Thomson 2004; Travis et al. 2025). This finding is consistent with the low plant specialisation, opportunistic foraging behaviour, and high population density of this species (Lowe et al. 2023). Manipulative experiments involving the exclusion or supplementation of honeybees are a complex task due to the widespread distribution and long foraging distances. As a result, we still lack a comprehensive synthesis to evaluate how common these processes are across different biogeographical regions, climates, and habitats.

Quantifying resource overlap of wild pollinators with the honeybee and their diet specialisation across ecological networks can provide a complementary approach to detect shifting preferences along a gradient of honeybee dominance (Cappellari et al. 2022). When floral resources are limited and pollinators compete asymmetrically, a low resource overlap and an altered foraging breadth should emerge, indicating potential competitive displacement where a superior competitor forces the inferior competitors to shift their niche and feed on sub-optimal floral resources (Magrach et al. 2017; Pasquali et al. 2025). Such displacement can reduce the abundance of the inferior competitor (Lindström et al. 2016; D. M. Thomson 2016), or, in the most extreme cases, lead to their local disappearance (Lázaro et al. 2021). In addition, other wild pollinators, such as bumblebees can also displace the honeybee, which, under certain circumstances, can be an inferior competitor (Balfour, Gandy, and Ratnieks 2015; Wignall et al. 2020). Foraging shifts are expected to increase with honeybee density and floral resource availability in the network (Gómez-Martínez et al. 2022). Low plant diversity can increase resource overlap between honeybees and other pollinators due to the lack of alternative floral resources, forcing species to forage on the same plants (Bernhardsson et al. 2024; Page et al. 2024), while functional trait similarity (e.g., body size and tongue length) can also reshape resource use (Cappellari et al. 2022). Insect pollinators are highly diverse, with Hymenoptera, Diptera, Lepidoptera and Coleoptera representing the key flower visiting orders, which differ in life histories, nesting requirements, diet preferences, and floral resource specialisation. Yet most research has focused on the relationship between honeybees and wild bees (Iwasaki and Hogendoorn 2022) with limited understanding of the responses of other key pollinator groups to increased honeybee dominance.

Using the European plant-pollinator network database (Lanuza et al. 2025), this study provides the first continental synthesis of the effect of honeybee dominance on resource overlap and diet specialisation across more than 1800 species of wild pollinators, including wild bees, hoverflies, butterflies and coleopterans. Based on more than 649 independent networks and 1625 sampling events, the role of the honeybee across European plant-pollinator networks was assessed in terms of visitation rate patterns and floral resource use. It was tested whether increasing dominance of the honeybee in the networks broadened foraging breadth of co-occurring wild pollinators and reduced their resource overlap with the honeybee, indicating a potential intensification of competition for floral resources. By using recently-developed scale-location models, we also explored the variability in pollinator response. These relationships were examined across the major biogeographical, climatic, and land-use gradients in Europe.

MATERIALS AND METHODS

Data

This work uses the recently compiled database of European plant-pollinator networks (EuP-PolNet; Lanuza et al. 2025). The harmonised taxonomy across the different networks allowed for testing resource overlap and diet specialisation at the species level across multiple networks. Weighted networks that met the following criteria were included: i) presence of honeybees, ii) more than one plant species, and iii) more than one pollinator species. The pollinator species were pooled into six non-overlapping taxonomic groups that are expected to interact differently with the honeybee due to distinct behavioural and morphological traits: wild bees (hymenopteran families: Apidae, Halictidae, Andrenidae, Colletidae, Megachilidae, Melittidae), non-bee hymenopterans (all families excluding wild bees), lepidopterans, coleopterans, hoverflies (dipteran family: Syrphidae), and non-syrphid flies (all dipteran families excluding Syrphidae). The final filtered dataset comprised 592 independent networks from 46 studies with a total of 1634 sampling events.

Resource overlap and diet specialisation

For each sampling event per network, we used the observed interactions to compute the Morisita-Horn index (Horn 1966) to quantify resource overlap between each focal pollinator species and the honeybee:

$$\text{Morisita-Horn} = \frac{2 \sum_k^S p_k h_k}{\left(\sum_k^S \left(\frac{p_k^2}{P^2} \right) + \sum_k^S \left(\frac{h_k^2}{H^2} \right) \right) PH}$$

Here, p_k is the number of interactions of the focal pollinator species with each plant species k and h_k is the number of interactions of the honeybee with each plant k . S is the number of plant species, P the total number of interactions of the focal pollinator, and H is the total number of interactions of the honeybee. This symmetric index ranges from 0 (no shared use

of resources) to 1 (identical resource use profile). The index normalises the visits by each species' total, using the relative distributions of resource use. Hence, it is not mathematically related to the abundance of focal pollinators or honeybees. The Müller index was not used as it is directional and mathematically related to honeybee dominance, ranging from 0 when honeybees are rare to 1 when they dominate the network, i.e. irrespective of the resource overlap, if honeybees are rare visitors overall, observed Müller index values will tend to 0, while if honeybees numerically dominate the network, Müller index values will inflate toward 1.

For each pollinator, we also estimated two complementary measures of diet specialisation, normalised degree and PDI (Poisot et al. 2012). Normalised degree is defined as $\frac{k_i}{N_{\text{plant}}}$, where k is the species degree (number of plant species visited by pollinator i) and N_{plant} is the total number of plant species in the network. As normalised degree is mathematically constrained by the number of plant species in a network, values are not directly comparable among networks with different plant species richness.

The pair difference index (PDI) was computed using its normalised version. For a species with interaction strengths $w_1 \geq w_2 \geq \dots \geq w_k$ across k plants, we set $p_i = \frac{w_i}{\max(w)}$ so that $p_1 = 1$. The index is defined as:

$$PDI_{\text{norm}} = \frac{\sum_i^k (1 - p_i)}{k - 1}$$

The index lies between 0 and 1, with 0 indicates a perfect generalist (all links equal), and 1 a perfect specialist (only one link). The Kullback–Leibler distance (d' ; Blüthgen, Menzel, and Blüthgen 2006) was not used, as in the absence of independent data on plant abundance, the metric d' is sensitive to the behaviour of the other pollinator species in the network, and the metric becomes mathematically related to the abundance of any dominant species (Poisot et al. 2012).

Null models

Most node-level metrics computed across networks can be affected by overall network topology. Null models can be effectively used to compare metrics across networks, removing the effect of neutral processes (Blüthgen and Staab 2024). For each network and sampling event, we created 500 null models using the Patefield algorithm (Patefield 1981). This null model maintains the total interactions of both pollinator and plant species constant. We used this null model because the total number of interactions per pollinator reflects the local abundance of honeybees and other pollinators in the habitat, while keeping the total number of interactions per plant constant preserves the fact that some plant species are inherently more attractive to pollinators than others. Keeping these constraints, the null model randomly reshuffled the pollinator visits on different plants. For each species, mean and SD from the 500 null models were computed for all metrics and used to compute standardised effect sizes: $SES = \frac{(\text{observed} - \text{mean}_{\text{null}})}{SD_{\text{null}}}$. Due to the generating constraints of the Patefield algorithm, in

some networks, SESs could not be computed because SD approximated 0. In further analyses, networks with $SD = 0$ or $IQR = 0$ were removed as SES could not be defined.

Drivers

In the EuPPollNet database each network contains information on geographical coordinates, sampled habitat, date of the sampling, and biogeographical regions. Summer mean temperature was derived by averaging the monthly averages from May to September for each sampling site (i.e., network) using the Chelsea database (Karger et al. 2023). Since there was a strong collinearity between climate and habitats, the original habitat types were grouped in two broad categories: (i) natural habitats including forest/woodland, riparian vegetation, semi-natural grasslands, moors and heathland, sclerophyllous vegetation, montane to alpine grasslands, beaches, dunes, sands, riparian; and (ii) anthropogenic habitats including intensive grasslands, agricultural land, agricultural margins, green urban areas, and ruderal vegetation. At the network level and for each sampling event, we computed the total number of plant partners, the total number of pollinator species, and the honeybee dominance as the ln-ratio between total honeybee interactions and non-honeybee interactions. The ln-ratio is 0 when honeybee interactions are equal to the sum of the interactions of all other pollinators sampled. We could not use the total abundance of honeybees, since the sampling effort differed between studies. Although the database includes information on floral abundance, inconsistent sampling methodologies limited their use, and only 74% of the studies reported such data.

Data analysis

Resource overlap with the honeybee and foraging breadth of wild pollinators

The three metrics (i.e., Morisita-Horn index, normalized degree, and PDI) were transformed into standardised effect sizes (SESs) and used as response variables in generalised linear mixed models (GLMMs) with a Gaussian distribution in the glmmTMB package (McGillcuddy et al. 2025). For each response, a location-scale model was fitted. Although ecological data often exhibit non-constant variance, this pattern is commonly considered noise that violates the assumptions of statistical models, while heteroscedasticity can often represent meaningful ecological processes (Cleasby and Nakagawa 2011). The location component (conditional in glmmTMB) tested the effect of pollinator group, honeybee dominance, number of plant partners, temperature, habitat type on the SES mean of each metric. A quadratic term for dominance was also tested, and the term was removed when linearity could not be rejected ($P > 0.05$). Two interactions were also tested: honeybee dominance with pollinator taxonomic groups and honeybee dominance with floral richness. The first tested whether the six pollinator taxonomic groups responded differently to the honeybees, while the second tested whether the availability of floral partners modified the honeybee dominance effect on wild pollinators. Sampling date nested in network ID, nested in Study ID, was used as a random structure. Second, a scale component (dispformula in glmmTMB) was used to test whether the number of floral partners and honeybee dominance affected the variance of the SES residuals. The same random structure was included in the scale component, but it was fitted independently from the conditional, as correlated random effects are not implemented in glmmTMB. Models

with and without the scale component were compared with the likelihood ratio test using AIC. In all models, LRTs always supported the inclusion of the two predictors and the random structure in the scale component. We tested for potential collinearity between predictors using a global model with no interactions using the `check_collinearity()` function in the performance package (Lüdecke et al. 2021). All predictors included in the modelling exhibited low VIFs (< 1.5). Continuous predictors were scaled using the `scale()` function. Model diagnostics were performed using the DHARMA package (Hartig 2016). All models performed well except for Morisita-Horn, which presented a slight deviation (Supplementary Information). All analyses were run using R version 4.5.1.

Sensitivity analysis

As resource overlap and diet specialisation metrics could be influenced by species degree and abundance, a sensitivity analysis was conducted, where focal pollinator species were removed based on their total number of individuals per network and sampling event. Main text analyses included only pollinator species with at least five individuals and a minimum of five honeybees per network. In the Supplementary Information, we presented the same models with different thresholds of minimum number of individuals per network per sampling event (min = 2, 3, 5, and 10). Results were qualitatively similar and showed strong consistency in the observed patterns.

RESULTS

General results

Honeybees visited approximately 50% of all plant species in the dataset and contributed on average 36% of the interactions per network (**Fig. S1-S2**), ranging from networks where they were present only sporadically and at low density (under 1% of visits) to networks where they were consistently the main visitor (up to 98% of visits). Hence, the remaining 50% of plant species were visited by other pollinators but not by the honeybee. Only 3.5% of plants were visited exclusively by the honeybee. As expected, there was a positive linear association between floral richness and pollinator richness across networks (**Fig. 1**). Both floral and pollinator species richness exhibited a weak correlation with honeybee normalised degree, with a trend for higher normalised degree at low species richness values. In the sampled networks, honeybee dominance was weakly negatively correlated with floral and pollinator richness and positively with honeybee normalised degree.

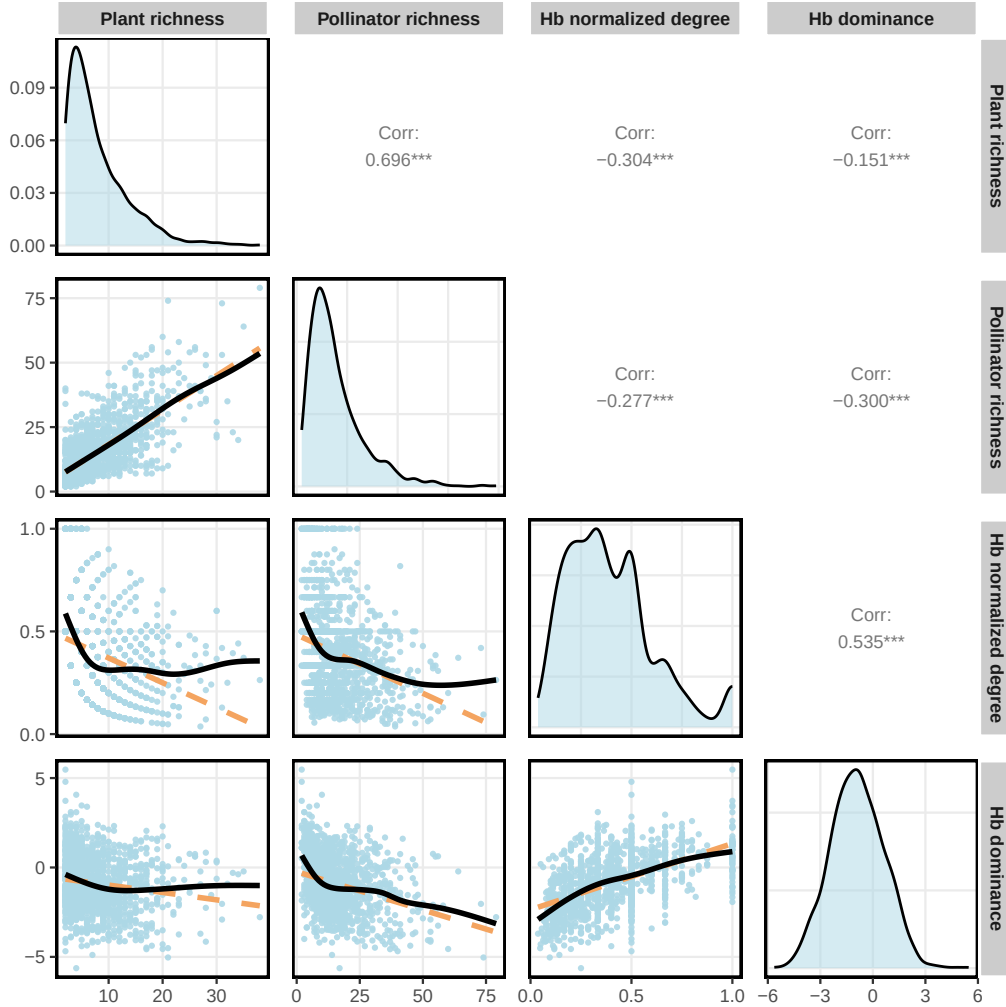


Figure 1. Scatterplots showing the relationships between the two components of network size (i.e., plant and pollinator species richness) and honeybee normalized degree and dominance (log ratio of honeybee interactions over total interactions). The dashed orange line represents the linear regression, and the solid black line represents the GAM fitted with cubic regression splines. Each point corresponds to an independent sampling event within a network. Only networks with at least five honeybee individuals were included. Diagonal panels represent variable density distributions, while upper panels indicate Pearson correlation coefficients between metrics.

207 *Resource overlap*

208 The standardized effect sizes of Morisita-Horn were mainly negative, indicating that in most
 209 cases, wild pollinators co-occurring with the honeybee exhibited a lower resource overlap than
 210 expected from the null models. The location component on the mean SES supported an

interaction between pollinator group and honey bee dominance (**Table S1a**), with a steeper slope for coleopterans and syrphids compared to other groups (**Fig. 2a**). The interaction between the number of plants and dominance was not strongly supported, while the number of plants available in the network reduced resource overlap. No effect of temperature or habitat was found. The scale component indicated an increase in the SES residual variance with dominance and a decrease with floral richness (**Table S1b**), meaning that pollinator resource overlap became more variable at high dominance and low floral richness.

Diet specialization

The SESs for the paired difference index (PDI) were mostly positive or around 0, indicating that pollinator specialization tended to be relatively higher than or similar to the expected values from the null models. The location-scale component indicated that mean PDI declined non-linearly with increasing honeybee dominance for all groups, except for the coleopterans which showed an opposite pattern (**Fig. 2B**). Coleopterans and lepidopterans were those with higher specialisation. A weaker interaction was found between dominance and floral richness on mean PDI, with a slightly steeper negative slope in networks with a higher number of flowering species. Networks sampled in warmer regions presented higher pollinator specialisation (**Table S2a**). The dispersion component indicated a negative effect of floral richness on the PDI residual variability, while dominance did not have an effect (**Table S2b**).

The SESs of normalised degree were mainly negative or around 0, indicating that the pollinator use of the available plants tended to be relatively lower than or similar to the expected values from the null models. Consistent with PDI, an interaction between taxonomic group and dominance was found (**Table S3a**), with normalised degree increasing with honeybee dominance in all groups except for coleopterans, which exhibited an opposite pattern (**Fig. 2C**). An interaction between floral richness and dominance was also found, where mean normalised degree declined with increasing floral richness but only at low dominance values. The dispersion component indicated an increase in the SES residual variance of normalised degree with dominance and a decrease with floral richness (**Table S3b**).

Model performance and sensitivity

The residuals' diagnostic of the four scale-location models was satisfactory considering the complexity of the data and the modelling. The deviations were minor, but statistically significant due to the large number of observations. The sensitivity analysis indicated that removing species with different thresholds of minimum interactions per network ($n = 2, 3, 4, 10$) did not change the observed patterns. Results of these sensitivity analyses are reported in Supplementary materials.

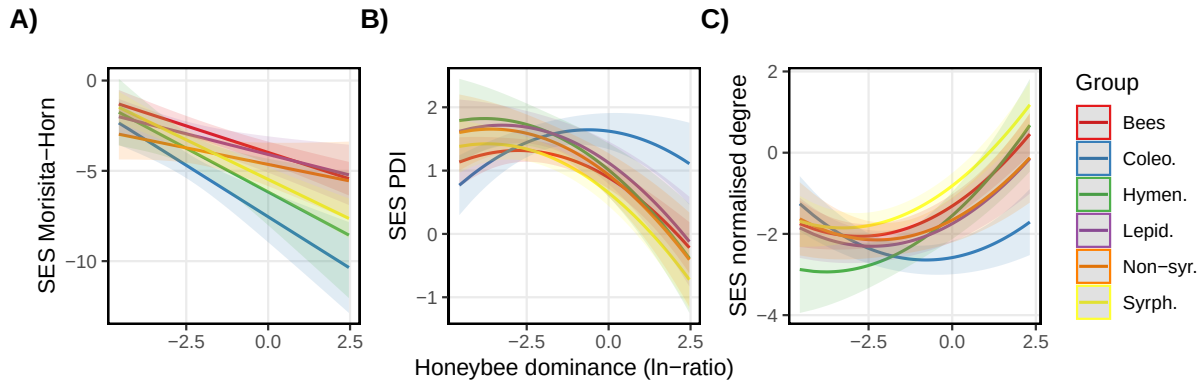


Figure 2. Effect of honeybee dominance on the standardised effect size (SES) of A) resource overlap with the honeybee (Morisita-Horn index), B) Paired difference index (PDI) and C) normalised degree of wild pollinators across different groups of flower-visiting insects. Lines represent predicted values from location-scale glmm. The analysis included networks with at least 5 honeybee individuals and wild pollinator species with at least 5 individuals per sampling event.

DISCUSSION

Honeybees were dominant visitors across European plant-pollinator networks, and their dominance was associated with a systematic decline in floral resource overlap with co-occurring wild pollinators, as well as a broader and more variable foraging breadth of co-occurring species. Although the observed associations were consistent across natural and disturbed habitats and climatic gradients, the major pollinator groups did not respond equally to the dominance of honeybees. As honeybees are known to remove a large share of floral resources, often from the most abundant and rewarding flowering species (Hung et al. 2018), this finding suggests that honeybees can reshape resource use of co-occurring wild pollinator species (Magrach, Tobajas, and Martin 2025). How important these diet shifts are for population dynamics, community diversity changes, or ecosystem functioning in the long-term remains an open question (Beaurepaire et al. 2025; Pascual Tudanca, Debandi, and Vázquez 2024).

The observed low overlap of wild pollinators with the flowering species visited by the honeybee may reflect a direct response to resource depletion, but possibly also to deterrent scent cues or other indicators of honeybee presence (Stout and Goulson 2001). Although the mechanisms underpinning this avoidance behaviour are still not fully understood, the finding of a reduced resource use overlap between the honeybee and other wild bees has been reported in many experimental and observational studies (Cappellari et al. 2022; Page and Williams 2023; Magrach, Tobajas, and Martin 2025). Manipulative studies that have excluded honeybees have demonstrated that a reduced resource overlap is often linked to short-term foraging shifts that can be reversed when honeybees are removed from the community (Pasquali et al.

266 2025). This pattern was also affected by plant diversity, where increasing plant niche space
267 generally reduces the probability of intra-species contact providing more foraging opportunities
268 to co-occurring species (Gómez-Martínez et al. 2022). The inclusion of a large number of
269 networks with a harmonised taxonomy allowed us to test whether this reduced overlap changed
270 for the major European pollinator groups. On the one hand, wild bees and lepidopterans
271 exhibited a relatively higher overlap with the honeybee than other groups, consistent with
272 the similar nectar requirements of these taxa. On the other hand, coleopterans exhibited
273 the lowest overlap. Observed coleopterans, such as Scarabaeidae (Cetoniinae), Oedemeridae,
274 Nitidulidae, and Melyridae, are often florivores that, besides removing pollen and nectar,
275 frequently damage flower parts, thereby reducing flower attractiveness for pollinators (Haas
276 and Lortie 2020). Coleopterans usually select flower species mainly based on a more extended
277 longevity, spending a long time feeding on the same species (Camurça et al. 2024). Hence, it
278 is likely that the observed reduced overlap was a consequence of honeybees avoiding flowers
279 either damaged or occupied by florivores (Boaventura et al. 2022).

280 The scale-location model also indicated that not only the mean resource overlap was affected
281 by the honeybee, but also its variability. This contrasts with many ecological and behavioural
282 studies where variability is often assumed to be constant, potentially masking informative bio-
283 logical and ecological patterns in the data (Nakagawa et al. 2025). An increase in behavioural
284 variability in pollinators in the presence of a dominant pollinator is consistent with a trait-
285 mediated shift in behaviour, i.e. species differed in their ability to avoid honeybee-used plants
286 (e.g. tongue length, body size, handling time, floral preferences) and under strong honeybee
287 pressure, some species shifted away (low overlap), while others could not avoid or even track
288 the same rewarding resources due to their trait similarity with the honeybee (Cappellari et al.
289 2022).

290 Optimal foraging theory also suggests that when resources are abundant and diverse and
291 species are not competing strongly, pollinators should concentrate on a few most-preferred
292 plants and have narrow food niches, while increased competition usually forces inferior com-
293 petitors to become more generalist in their resource use (MacArthur and Pianka 1966). After
294 removing the effect of neutral processes using null models, we observed a general decrease in
295 diet specialisation with increasing honeybee dominance for all pollinator taxonomic groups,
296 except for florivorous coleopterans. Again, pollinator taxonomic groups showed contrasting
297 responses likely due to trait-mediated differences among species in their ability to adjust their
298 foraging behaviour under increased honeybee dominance. An interaction between floral di-
299 versity and honeybee dominance was also found, with a more pronounced specialisation with
300 increasing floral diversity only at low dominance, while at high dominance levels, specialisation
301 did not depend on floral diversity. Again, the location scale model also detected an increase
302 in the variability of responses among species, suggesting that honeybee pressures might am-
303 plify intra- and interspecific differences, likely driven by morphological and behavioural traits
304 affecting resource use (Cappellari et al. 2022).

305 Although species can adjust to competition by shifting their diets, the coexistence of a super-
306 abundant pollinator such as the honeybee and native pollinators can be fragile and will only

in part depend on floral resource availability. It is important to stress that asymmetric competition among pollinators is probably the rule across plant-pollinator networks (Sponsler and Bratman 2021), and it has been observed in many other pollinator species besides the honeybee (Inouye 1978; Taggar, McGrath, and Despland 2021). However, several studies examining nectar and pollen landscapes have indicated that honeybee-induced diet shifts are often associated with poorer nutritional quality in wild pollinator diets (Page et al. 2024; Wignall et al. 2020). Hence, the effectiveness of conservation interventions for wild pollinators, in particular across agricultural landscapes, should always consider the potential competition with the honeybees (Bommarco et al. 2021; Raderschall et al. 2022). At the same time, maintaining honeybee populations across agricultural landscapes is crucial for delivering adequate crop pollination. Under these circumstances, it is vital to identify the minimum honeybee hive placement to balance yield and biodiversity by providing needed pollination for the crop while minimising competition with wild pollinators (Fijen, Bodegraven, and Lucassen 2022). From a wild plant pollination perspective, the consequences of these diet shifts are difficult to predict and are expected to be strongly context-dependent (Gao et al. 2025; Magrach et al. 2017; Pascual Tudanca, Debandi, and Vázquez 2024; Valido, Rodríguez-Rodríguez, and Jordano 2019). Although managed honey bees per se may not be a conservation priority, beekeeping is profoundly rooted in the human-nature history (Sponsler and Bratman 2021) and is associated with key cultural services, highlighting the need to co-create effective and context-relevant strategies to reconcile beekeeping and the conservation of wild pollinators.

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SUPPLEMENTARY INFORMATION

Table S1. Scale-location model testing multiple predictors on the standardized effect size of pairwise resource overlap (Morisita–Horn) between the honeybee and wild pollinator species across European networks. The conditional component tested the effects of group, honeybee dominance (linear and quadratic terms), temperature, habitat, plant richness, and the interactions between dominance and group and between dominance and plant richness on mean PDI. The dispersion component tested the effects of dominance and plant richness on the residual variance. Both components included pollinator species as a random intercept. Each observation corresponded to a species with at least five individuals recorded in a sampling event that also contained at least five honeybees.

Term	Estimate	SE	z	p
(a) Conditional (mean)				
(Intercept)	-2.807	0.278	-10.099	0.000
GroupColeo.	-2.955	0.471	-6.273	0.000
GroupHymen.	-1.770	0.587	-3.017	0.003
GroupLepid.	-0.252	0.341	-0.741	0.459
GroupNon-syr.	-0.904	0.407	-2.224	0.026
GroupSyrph.	-1.175	0.191	-6.168	0.000
scale(Dominance)	-0.822	0.136	-6.029	0.000
scale(Plant)	-0.218	0.123	-1.766	0.077
HabitatNatural	-0.531	0.334	-1.587	0.113
scale(Temperature)	-0.293	0.176	-1.667	0.096
GroupColeo.:scale(Dominance)	-0.771	0.352	-2.193	0.028
GroupHymen.:scale(Dominance)	-0.531	0.504	-1.053	0.292
GroupLepid.:scale(Dominance)	0.182	0.219	0.830	0.407
GroupNon-syr.:scale(Dominance)	0.309	0.321	0.962	0.336
GroupSyrph.:scale(Dominance)	-0.403	0.158	-2.548	0.011
(b) Dispersion (variance)				
(Intercept)	1.241	0.075	16.569	0.000
scale(Dominance)	0.360	0.039	9.197	0.000
scale(Plant)	-0.144	0.045	-3.171	0.002

Table S2. Scale-location model testing multiple predictors on the standardized effect size of paired difference index (PDI) between the honeybee and wild pollinator species across European networks. The conditional component tested the effects of group, honeybee dominance (linear and quadratic terms), temperature, habitat, plant richness, and the interactions between dominance and group and between dominance and plant richness on mean PDI. The dispersion component tested the effects of dominance and plant richness on the residual variance. Both components included pollinator species as a random intercept. Each observation corresponded to a species with at least five individuals recorded in a sampling event that also contained at least five honeybees.

Term	Estimate	SE	z	p
(a) Conditional (mean)				
(Intercept)	1.082	0.083	13.068	0.000
GroupColeo.	0.470	0.102	4.594	0.000
GroupHymen.	0.249	0.147	1.696	0.090
GroupLepid.	0.294	0.115	2.554	0.011
GroupNon-syr.	0.144	0.119	1.203	0.229
GroupSyrph.	-0.123	0.061	-2.013	0.044
scale(Dominance)	-0.410	0.057	-7.183	0.000
HabitatNatural	-0.028	0.103	-0.270	0.787
scale(Temperature)	0.121	0.051	2.359	0.018
scale(log(Plant))	-0.016	0.038	-0.422	0.673
scale(I(Dominance ²))	-0.220	0.056	-3.952	0.000
GroupColeo.:scale(Dominance)	0.340	0.090	3.764	0.000
GroupHymen.:scale(Dominance)	-0.165	0.132	-1.252	0.211
GroupLepid.:scale(Dominance)	-0.078	0.098	-0.797	0.426
GroupNon-syr.:scale(Dominance)	-0.132	0.115	-1.148	0.251
GroupSyrph.:scale(Dominance)	-0.151	0.057	-2.638	0.008
scale(Dominance):scale(log(Plant))	-0.095	0.033	-2.859	0.004
(b) Dispersion (variance)				
(Intercept)	0.143	0.030	4.751	0.000
scale(Dominance)	0.054	0.028	1.910	0.056
scale(log(Plant))	-0.033	0.028	-1.175	0.240
scale(Dominance):scale(log(Plant))	0.111	0.028	3.958	0.000

Table S3. Scale-location model testing multiple predictors on the standardized effect size of normalised degree between the honeybee and wild pollinator species across European networks. The conditional component tested the effects of group, honeybee dominance (linear and quadratic terms), temperature, habitat, plant richness, and the interactions between dominance and group and between dominance and plant richness on mean PDI. The dispersion component tested the effects of dominance and plant richness on the residual variance. Both components included pollinator species as a random intercept. Each observation corresponded to a species with at least five individuals recorded in a sampling event that also contained at least five honeybees.

Term	Estimate	SE	z	p
(a) Conditional (mean)				
(Intercept)	-1.521	0.140	-10.828	0.000
GroupColeo.	-0.836	0.136	-6.155	0.000
GroupHymen.	-0.455	0.198	-2.294	0.022
GroupLepid.	-0.352	0.146	-2.414	0.016
GroupNon-syr.	-0.230	0.180	-1.281	0.200
GroupSyrph.	0.390	0.085	4.577	0.000
scale(Dominance)	0.736	0.072	10.190	0.000
HabitatNatural	-0.037	0.159	-0.234	0.815
scale(Temperature)	-0.095	0.086	-1.105	0.269
scale(log(Plant))	-0.168	0.052	-3.229	0.001
scale(I(Dominance ²))	0.385	0.073	5.276	0.000
GroupColeo.:scale(Dominance)	-0.546	0.119	-4.576	0.000
GroupHymen.:scale(Dominance)	0.277	0.199	1.393	0.164
GroupLepid.:scale(Dominance)	-0.103	0.132	-0.784	0.433
GroupNon-syr.:scale(Dominance)	-0.144	0.179	-0.804	0.422
GroupSyrph.:scale(Dominance)	0.137	0.081	1.676	0.094
scale(Dominance):scale(log(Plant))	0.097	0.044	2.221	0.026
(b) Dispersion (variance)				
(Intercept)	0.335	0.021	15.879	0.000
scale(Dominance)	-0.010	0.020	-0.520	0.603
scale(log(Plant))	0.086	0.019	4.432	0.000

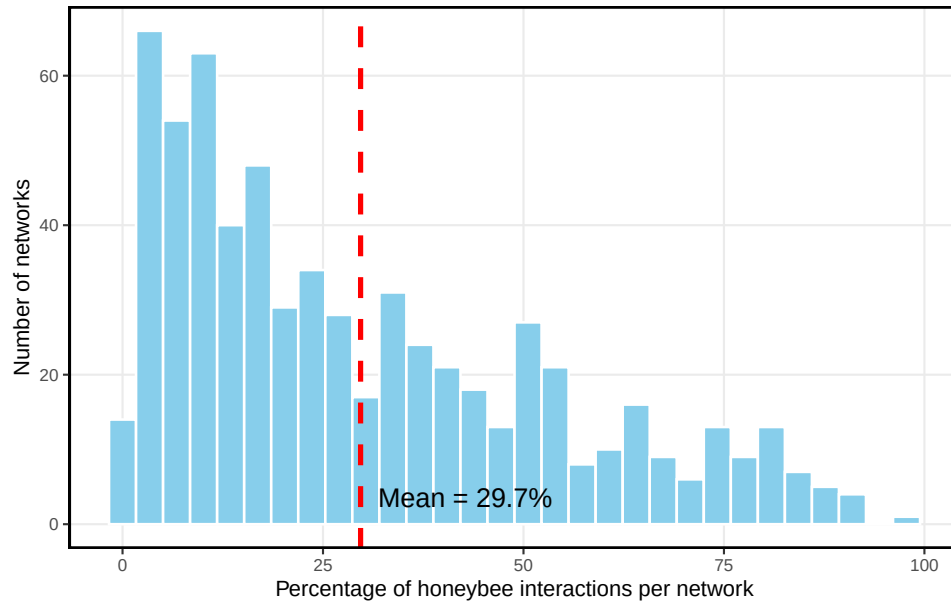


Figure S1. Histogram showing the mean proportion of interactions of honeybees per network. The average value across all networks is indicated with a vertical dashed red bar.

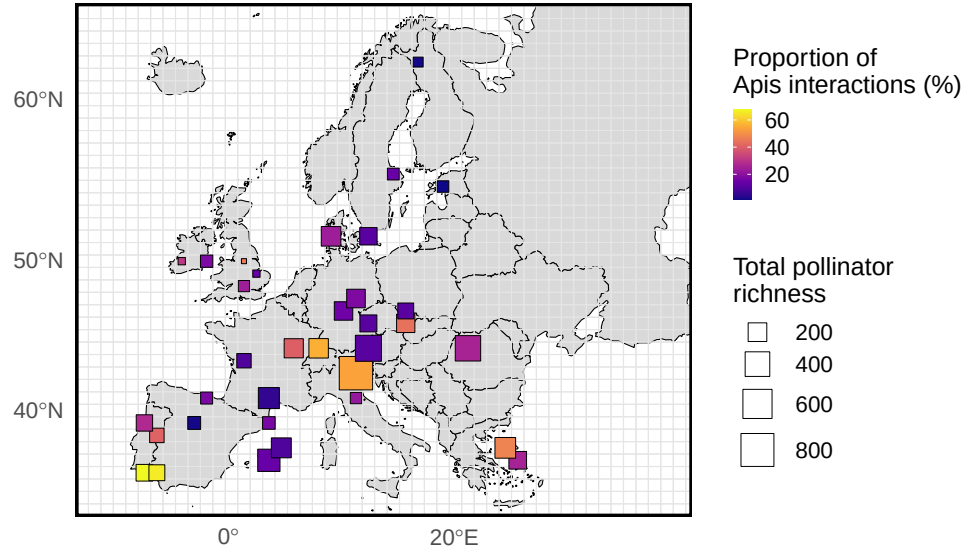


Figure S2. Approximate locations of the studies, showing the proportion of *Apis* interactions weighted by the total number of pollinator species per study.